AMA572 Lecture 3.

First Passage Times and Black-Cox Model

1 Laplace Transforms of First Passage Times

Instead of using the probability (Girsanov) transform, another way to compute the first passage times is to use Laplace transform via martingales. Consider the first passage time

$$\tau_b = \min\{t \geq 0 : \ W_{\mu,\sigma}(t) = b\}, \ \ b > 0.$$

Here, without loss of generality,we assume $b > 0$, and $W_{\mu,\sigma}(t)$ is a Brownian motion with drift μ and volatility σ, i.e., $W_{\mu,\sigma}(t) = \mu t + \sigma W(t)$. Note that $W_{\mu,\sigma}(\tau_b) = b$ on the set $\tau_b < \infty$.

Theorem. For any $s > 0$, the Laplace transform of τ_b is given by

$$E[e^{-s\tau_b}] = \exp\left\{-\frac{b}{\sigma^2}\left(\sqrt{\mu^2 + 2\sigma^2 s} - \mu\right)\right\}, \tag{1}$$

$$P\left(\tau_b < \infty\right) = \begin{cases} \exp\left(2\mu b/\sigma^2\right) < 1, & \text{if } \mu b < 0, \\ 1, & \text{otherwise.} \end{cases} \tag{2}$$

Proof. Let $s = \theta\mu + \frac{1}{2}\theta^2\sigma^2$, for any numbers $\theta > 0$ and $s > 0$. More precisely, since θ is a positive solution of a quadratic equation we have

$$\theta = \frac{1}{\sigma^2}\left(\sqrt{\mu^2 + 2\sigma^2 s} - \mu\right) > 0.$$

Thus, for any $s > 0$, we can always define $\theta > 0$ as above.

It is easy to see that

$$\exp\{\theta W_{\mu,\sigma}(t) - st\}$$

is a martingale. Therefore, by the optional sampling theorem, for any $t > 0$,

$$1 = E[\exp\{\theta W_{\mu,\sigma}(t \wedge \tau_b) - s(\tau_b \wedge t)\}]. \tag{3}$$

Since $s > 0$ and $\theta > 0$,

$$\begin{aligned} 0 \ &\leq \ \exp\{\theta W_{\mu,\sigma}(t \wedge \tau_b) - s(\tau_b \wedge t)\} \\ &\leq \ \exp\{\theta b - s(\tau_b \wedge t)\} \leq \exp\{\theta b\}, \end{aligned}$$

independent of t. Therefore, by the dominated convergence theorem, we can let $t \to \infty$ in (3) to get

$$\begin{aligned} 1 \ &= \ E\left[\lim_{t\to\infty}\left(\exp\{\theta W_{\mu,\sigma}(t) - st\} I\{\tau_b = \infty\}\right)\right] \\ &\quad + E[\exp\{\theta W_{\mu,\sigma}(\tau_b) - s(\tau_b)\} \cdot I\{\tau_b < \infty\}] \\ &= \ E[\exp\{\theta W_{\mu,\sigma}(\tau_b) - s\tau_b\} \cdot I\{\tau_b < \infty\}]; \end{aligned}$$

note the first term is zero because limsup$W(t) < b$ on the set $\tau_b = \infty$ and, again, $\theta > 0$. Since $W_{\mu,\sigma}(\tau_b) = b$ on the set $\tau_b < \infty$, we further have

$$E[\exp\{\theta b - s\tau_b\}] = E[\exp\{\theta b - s\tau_b\} \cdot I\{\tau_b < \infty\}] = 1.$$

In summary, we have

$$E[e^{-s\tau_b}] = e^{-\theta b} = \exp\left\{-\frac{b}{\sigma^2}\left(\sqrt{\mu^2 + 2\sigma^2 s} - \mu\right)\right\}.$$

As a by-product, we also have

$$P\left(\tau_b < \infty\right) = \lim_{s \to 0} E[e^{-s\tau_b}] = \exp\{-b\left(\sqrt{\mu^2} - \mu\right)/\sigma^2\}.$$

Since $\sqrt{\mu^2} = |\mu|$, we get that

$$P\left(\tau_b < \infty\right) = \begin{cases} \exp\left(2\mu b/\sigma^2\right) < 1, & \text{if } \mu b < 0; \\ 1, & \text{otherwise.} \end{cases}$$

Generally speaking, one can only numerically invert the Laplace transform to compute the density function. However, in this special case, the Laplace transform can be inverted analytically.

The Laplace inversion table (available in many handbooks of mathematical tables) gives the Laplace inversion of the function $\exp\{-a\sqrt{s}\}$, $a > 0$, as

$$\frac{a}{2\sqrt{\pi}} \frac{1}{x^{3/2}} \exp\left\{-\frac{a^2}{4x}\right\}.$$

We can use this result to invert the Laplace transform in (1) to calculate the density of τ_b explicitely.

Lemma. The density of τ_b is given by

$$f_{\tau_b}(x) = \frac{b}{\sigma\sqrt{2\pi x^3}} \exp\left\{-\frac{(b - x\mu)^2}{2x\sigma^2}\right\}.$$

Proof. Using the Laplace inversion table, we have

$$\int_0^\infty e^{-sx} \cdot \frac{a}{2\sqrt{\pi}} \frac{1}{x^{3/2}} \exp\left\{-\frac{a^2}{4x}\right\} dx = \exp\{-a\sqrt{s}\},\ a > 0.$$

Thus, changing s to $\mu^2 + 2\sigma^2 s$, we get

$$\begin{aligned} &\int_0^\infty e^{-(\mu^2 + 2\sigma^2 s)x} \cdot \frac{a}{2\sqrt{\pi}} \frac{1}{x^{3/2}} \exp\left\{-\frac{a^2}{4x}\right\} dx \\ &= \exp\left\{-a\left(\sqrt{\mu^2 + 2\sigma^2 s}\right)\right\},\ a > 0, \end{aligned}$$

$$\begin{aligned}\Longrightarrow \qquad & \int_0^\infty e^{-2\sigma^2 sx}\cdot\frac{a}{2\sqrt{\pi}}\frac{1}{x^{3/2}}\exp\left\{-\frac{a^2}{4x}-\mu^2 x\right\}dx \\ & =\exp\left\{-a\left(\sqrt{\mu^2+2\sigma^2 s}\right)\right\},\ a>0.\end{aligned}$$

Changing the variable and letting $y=2\sigma^2 x$ yield

$$\begin{aligned}& \int_0^\infty e^{-sy}\cdot\frac{1}{2\sigma^2}\frac{a}{2\sqrt{\pi}}\frac{2\sqrt{2}\sigma^3}{y^{3/2}}\exp\left\{-\frac{a^2\sigma^2}{2y}-\frac{\mu^2 y}{2\sigma^2}\right\}dy \\ & =\exp\left\{-a\left(\sqrt{\mu^2+2\sigma^2 s}\right)\right\},\ a>0,\end{aligned}$$

$$\begin{aligned}\Longrightarrow \qquad & \int_0^\infty e^{-sy}\cdot\frac{a}{\sqrt{2\pi y^3}}\sigma\exp\left\{-\frac{a^2\sigma^2}{2y}-\frac{\mu^2 y}{2\sigma^2}\right\}dy \\ & =\exp\left\{-a\left(\sqrt{\mu^2+2\sigma^2 s}\right)\right\},\ a>0.\end{aligned}$$

In summary, with $a=b/\sigma^2$ in the above equation and multiplying by the constant $\exp\{-\frac{b}{\sigma^2}(-\mu)\}$, we have

$$\begin{aligned}f_{\tau_b}(y) &= \frac{b}{\sigma\sqrt{2\pi y^3}}\exp\left\{-\frac{b^2}{2y\sigma^2}-\frac{\mu^2 y}{2\sigma^2}\right\}\cdot\exp\left\{-\frac{b}{\sigma^2}(-\mu)\right\} \\ &= \frac{b}{\sigma\sqrt{2\pi y^3}}\exp\left\{-\frac{b^2+\mu^2y^2-2by\mu}{2y\sigma^2}\right\} \\ &= \frac{b}{\sigma\sqrt{2\pi y^3}}\exp\left\{-\frac{(b-y\mu)^2}{2y\sigma^2}\right\}.\end{aligned}$$

The derivation of the Laplace transform only relies on finding a martingale, applying the optional sampling theorem, and the fact $W_{\mu,\sigma}(\tau_b)=b$. This makes it more suitable to solve more complied first passage time problems.

2 Basic Setting of the Black-Cox Model

Assume that the asset value of the firm V_t follows a geometric Brownian motion under a risk-adjusted probability:

$$\frac{dV_t}{V_t}=(r-\delta)dt+\sigma dW_t,$$

whose solution is given by

$$V_t=V_0\exp\left\{\left(r-\delta-\frac{1}{2}\sigma^2\right)t+\sigma W_t\right\},$$

where r is the risk-free interest rate, δ is the total payout rate the firm's investors (including both bond and equity holders, as bond coupon and stock dividend payments).

Now the firm issues a bond with par value 1, maturity T and coupon rate ρ with $\rho < \delta$. The default may happen at any time between 0 and T. In fact the default happens at the first time the process V_t falls below V_B:

$$\tau = \min\{t \geq 0 : V_t \leq V_B\}, \;\; V_B < V_0.$$

Upon default, the bondholder gets

$$ce^{-r(T-\tau)}, \; 0 \leq c \leq 1.$$

In other words, we assume the recovery at the fraction of otherwise identical treasury bond value.

In this setting, the payoff to the bond consists of three parts, the par value at maturity if no default, the recovery value upon default, and the continuous coupon payment up to $\tau \wedge T$. Therefore, the bond price at time zero is given by

$$B(V; V_B, T) = E\left[e^{-rT} I\{\tau > T\} + e^{-r\tau} \cdot ce^{-r(T-\tau)} I\{\tau \leq T\} + \int_0^{\tau \wedge T} \rho e^{-rs} ds\right].$$

3 Laplace Transform of the Bond Price

Let

$$\tau_b = \min\{t \geq 0 : \; W_{\mu,\sigma}(t) = b\}, \; b < 0.$$

It can be shown easily that

$$E[e^{-s\tau_b}] = \exp\left\{\frac{b}{\sigma^2}\left(\sqrt{\mu^2 + 2\sigma^2 s} + \mu\right)\right\}, \; b < 0.$$

Now in the Black-Cox setting,

$$\begin{aligned}
\tau &= \min\{t \geq 0 : V_t \leq V_B\} \\
&= \min\left\{t \geq 0 : V_0 \exp\{(r - \delta - \frac{1}{2}\sigma^2)t + \sigma W_t\} \leq V_B\right\} \\
&= \min\left\{t \geq 0 : (r - \delta - \frac{1}{2}\sigma^2)t + \sigma W_t \leq \log(V_B/V_0)\right\}.
\end{aligned}$$

Thus,

$$
\begin{aligned}
E[e^{-s\tau_b}] &= \exp\left\{\frac{\log(V_B/V_0)}{\sigma^2}\left(\sqrt{(r-\delta-\frac{1}{2}\sigma^2)^2+2\sigma^2 s}+(r-\delta-\frac{1}{2}\sigma^2)\right)\right\} \\
&= \exp\left\{\log(V_B/V_0)\cdot\left(\sqrt{(\frac{r-\delta}{\sigma^2}-\frac{1}{2})^2+\frac{2}{\sigma^2}s}+\left(\frac{r-\delta}{\sigma^2}-\frac{1}{2}\right)\right)\right\} \\
&= \exp\left\{\log(V_B/V_0)\cdot\left(\sqrt{\xi^2+\frac{2}{\sigma^2}s}+\xi\right)\right\},
\end{aligned}
$$

where the constant $\xi = \frac{r-\delta}{\sigma^2}-\frac{1}{2}$.

Thus, we have the Laplace transform for the bond price, except the recovery part, is given by

$$
\begin{aligned}
&\int_0^\infty e^{-sT}E\left[e^{-rT}I\{\tau>T\}+\int_0^{\tau\wedge T}\rho e^{-rs}ds\right]dT \\
&= E\left\{\int_0^\infty e^{-sT}\left[e^{-rT}I\{\tau>T\}+\frac{\rho}{r}\left(1-e^{-r(\tau\wedge T)}\right)\right]dT\right\} \\
&= E\left\{\int_0^\tau e^{-sT}e^{-rT}dT+\int_0^\tau e^{-sT}\frac{\rho}{r}\left(1-e^{-rT}\right)dT+\int_\tau^\infty e^{-sT}\frac{\rho}{r}\left(1-e^{-r\tau}\right)dT\right\} \\
&= E\left\{\left(1-\frac{\rho}{r}\right)\int_0^\tau e^{-(s+r)T}dT+\int_0^\infty e^{-sT}\frac{\rho}{r}dT-\frac{\rho}{r}e^{-r\tau}\int_\tau^\infty e^{-sT}dT\right\} \\
&= E\left\{\left(1-\frac{\rho}{r}\right)\frac{1-e^{-(s+r)\tau}}{s+r}+\frac{\rho}{r}\frac{1}{s}-\frac{\rho}{r}\frac{e^{-(s+r)\tau}}{s}\right\} \\
&= \frac{\rho+s}{s(r+s)}(1-E[e^{-(r+s)\tau}]),
\end{aligned}
$$

which the first equality follows by changing the order the integration. Now, we only need to compute the Laplace transform of the recovery part:

$$
\begin{aligned}
&\int_0^\infty e^{-sT}E\left[e^{-r\tau}\cdot ce^{-r(T-\tau)}I\{\tau\le T\}\right]dT \\
&= c\int_0^\infty e^{-(s+r)T}E\left[I\{\tau\le T\}\right]dT \\
&= cE\left\{\int_0^\infty e^{-(s+r)T}I\{\tau\le T\}dT\right\} \\
&= cE\left\{\int_\tau^\infty e^{-(s+r)T}dT\right\} \\
&= \frac{c}{s+r}E[e^{-(s+r)\tau}].
\end{aligned}
$$

In summary, the Laplace transform of the bond price is given by

$$\begin{aligned}
&\int_0^\infty e^{-sT} B(V; V_B, T) dT \\
&= \frac{\rho + s}{s(r+s)}(1 - E[e^{-(r+s)\tau}]) + \frac{c}{s+r} E[e^{-(s+r)\tau}] \\
&= \frac{\rho + s}{s(r+s)} + \left(\frac{c}{s+r} - \frac{\rho + s}{s(r+s)} \right) E[e^{-(s+r)\tau}] \\
&= \frac{\rho + s}{s(r+s)} - \left(\frac{\rho}{r}\frac{1}{s} + \left(1 - \frac{\rho}{r} - c\right) \frac{1}{r+s} \right) \\
&\quad \times \exp\left\{ \log(V_B/V_0) \cdot \left(\sqrt{\xi^2 + \frac{2}{\sigma^2}(s+r)} + \xi \right) \right\}.
\end{aligned}$$

Fortunately, the above Laplace transform can be inverted analytically; see a hwk problem.

The equity price E can be obtained simply by using the fact that the equity value plus the bond value minus the bankruptcy cost is equal to the asset value of the firm, as there is no tax in this model. We will have a more general model includes tax later when we discuss Leland's model. So we skip the equity value here.

Empirically, the Black-Cox model leads to better possibilities of credit spreads than Merton's model. For example, it generates better spreads with humped shapes, partly because the default can happen any time between 0 and T. However, several main issues remain, such as the model cannot produce non-zero credit spreads observed in practice for short maturity bonds, and cannot generate flexible credit spread curves for speculative bonds.

4 Practical Implementation of Black-Cox Model

See the menu by CreditGrades.

HWK.

Invert the Laplace transform to get a formula for the bond price $B(V; V_B, T)$. Use the following Laplace inversion table:

$$\text{The Laplace inversion of } \frac{\rho + s}{s(r+s)} \text{ is } \frac{r-\rho}{r} e^{-rT} + \frac{\rho}{r},$$

$$\text{the Laplace inversion of } \frac{1}{s-\beta} \exp\{-\sqrt{\alpha s}\},\ \beta > 0,\ \alpha > 0,$$

is given by

$$e^{\beta T} \cdot \left\{ e^{\sqrt{\alpha\beta}} \Phi\left(-\sqrt{\frac{\alpha}{2T}} - \sqrt{2\beta T} \right) + e^{-\sqrt{\alpha\beta}} \Phi\left(-\sqrt{\frac{\alpha}{2T}} + \sqrt{2\beta T} \right) \right\}.$$

1 Practical Implementation of the Black-Cox Model

Under the risk-neutral measure, the firm value follows the geometric Brownian motion

$$\frac{dV_t}{V_t} = (r - \delta)\, dt + \sigma\, dW_t,$$

with solution

$$V_t = V_0 \exp\left\{\left(r - \delta - \frac{1}{2}\sigma^2\right) t + \sigma W_t\right\}.$$

The firm defaults when its asset value first hits a lower default barrier V_B, where

$$0 < V_B < V_0.$$

Thus the default time is

$$\tau = \inf\{t \geq 0 : V_t \leq V_B\}.$$

Taking logarithms, default occurs when

$$\left(r - \delta - \frac{1}{2}\sigma^2\right) t + \sigma W_t \leq \log\left(\frac{V_B}{V_0}\right).$$

Since $V_B < V_0$,

$$\log\left(\frac{V_B}{V_0}\right) < 0.$$

Therefore, the Black-Cox default time is a first downward passage time of a Brownian motion with drift.

Define

$$\xi = \frac{r - \delta}{\sigma^2} - \frac{1}{2}.$$

Then the Laplace transform of the default time is

$$\mathbb{E}\left[e^{-s\tau}\right] = \exp\left\{\log\left(\frac{V_B}{V_0}\right)\left[\sqrt{\xi^2 + \frac{2s}{\sigma^2}} + \xi\right]\right\}.$$

1.1 Bond Payoff

Consider a bond with face value 1, maturity T, continuous coupon rate ρ, and recovery parameter $c \in [0, 1]$. The bond payoff has three components:

$$e^{-rT}\mathbf{1}_{\{\tau > T\}},$$

which is the discounted par payment if no default occurs before maturity;

$$e^{-r\tau} c e^{-r(T-\tau)}\mathbf{1}_{\{\tau \leq T\}},$$

which is the discounted recovery payment if default occurs before maturity; and

$$\int_0^{\tau \wedge T} \rho e^{-rs}\, ds,$$

which is the discounted continuous coupon payment until default or maturity.

Hence the time-zero bond price is

$$B(V_0; V_B, T) = \mathbb{E}\left[e^{-rT}\mathbf{1}_{\{\tau > T\}} + e^{-r\tau} c e^{-r(T-\tau)}\mathbf{1}_{\{\tau \le T\}} + \int_0^{\tau \wedge T} \rho e^{-rs}\, ds\right].$$

1.2 Why the Laplace Transform Is Useful

The direct valuation of the bond requires the distribution of the first-passage time τ. Instead of working directly with the density of τ, we use its Laplace transform:

$$\mathbb{E}\left[e^{-s\tau}\right].$$

This is useful because the first-passage-time Laplace transform has a closed-form expression, while the direct maturity-dependent pricing formula is more difficult to derive.

Taking the Laplace transform of the bond price with respect to maturity T, define

$$\widehat{B}(s) = \int_0^{\infty} e^{-sT} B(V_0; V_B, T)\, dT.$$

The Black-Cox model gives

$$\widehat{B}(s) = \frac{\rho + s}{s(r+s)} - \left[\frac{\rho}{r}\frac{1}{s} + \left(1 - \frac{\rho}{r} - c\right)\frac{1}{r+s}\right]\mathbb{E}\left[e^{-(s+r)\tau}\right].$$

Substituting the default-time transform,

$$\mathbb{E}\left[e^{-(s+r)\tau}\right] = \exp\left\{\log\left(\frac{V_B}{V_0}\right)\left[\sqrt{\xi^2 + \frac{2(s+r)}{\sigma^2}} + \xi\right]\right\}.$$

Let

$$q = \frac{V_B}{V_0} \in (0,1), \qquad L = \log\left(\frac{V_0}{V_B}\right) > 0.$$

Then

$$\log q = -L.$$

Therefore,

$$\mathbb{E}\left[e^{-(s+r)\tau}\right] = q^{\xi} \exp\left\{-L\sqrt{\xi^2 + \frac{2(s+r)}{\sigma^2}}\right\}.$$

Now,

$$\xi^2 + \frac{2(s+r)}{\sigma^2} = \frac{2}{\sigma^2}\left(s + r + \frac{1}{2}\sigma^2\xi^2\right).$$

Define

$$\lambda = r + \frac{1}{2}\sigma^2\xi^2,$$

and

$$\alpha = \frac{2L^2}{\sigma^2}.$$

Then

$$L\sqrt{\xi^2 + \frac{2(s+r)}{\sigma^2}} = \sqrt{\alpha(s+\lambda)}.$$

Hence

$$\mathbb{E}\left[e^{-(s+r)\tau}\right] = q^{\xi}\exp\left\{-\sqrt{\alpha(s+\lambda)}\right\}.$$

Thus,

$$\widehat{B}(s) = \frac{\rho+s}{s(r+s)} - q^{\xi}\left[\frac{\rho}{r}\frac{1}{s} + \left(1 - \frac{\rho}{r} - c\right)\frac{1}{r+s}\right]\exp\left\{-\sqrt{\alpha(s+\lambda)}\right\}.$$

1.3 Laplace Inversion

The first term has inverse

$$\mathcal{L}^{-1}\left[\frac{\rho+s}{s(r+s)}\right](T) = \frac{r-\rho}{r}e^{-rT} + \frac{\rho}{r}.$$

This is the default-free continuous-coupon bond price:

$$B_{\mathrm{rf}}(T) = e^{-rT} + \int_0^T \rho e^{-rt}\,dt = \frac{\rho}{r} + \frac{r-\rho}{r}e^{-rT}.$$

Next, define

$$G_a(T) = \mathcal{L}^{-1}\left[\frac{1}{s+a}\exp\left\{-\sqrt{\alpha(s+\lambda)}\right\}\right](T).$$

Set

$$u = s + \lambda.$$

Then

$$s + a = u - (\lambda - a).$$

Define

$$\beta_a = \lambda - a.$$

Therefore,

$$\frac{1}{s+a}\exp\left\{-\sqrt{\alpha(s+\lambda)}\right\} = \frac{1}{u-\beta_a}\exp\{-\sqrt{\alpha u}\}.$$

Using the Laplace inversion formula,

$$\mathcal{L}^{-1}\left[\frac{1}{s-\beta}\exp\{-\sqrt{\alpha s}\}\right](T) = e^{\beta T}\left[e^{\sqrt{\alpha\beta}}\Phi\left(-\sqrt{\frac{\alpha}{2T}}-\sqrt{2\beta T}\right)+e^{-\sqrt{\alpha\beta}}\Phi\left(-\sqrt{\frac{\alpha}{2T}}+\sqrt{2\beta T}\right)\right],$$

we obtain the shifted inverse

$$G_a(T) = e^{-aT}\left[e^{\sqrt{\alpha(\lambda-a)}}\Phi\left(-\sqrt{\frac{\alpha}{2T}}-\sqrt{2(\lambda-a)T}\right)+e^{-\sqrt{\alpha(\lambda-a)}}\Phi\left(-\sqrt{\frac{\alpha}{2T}}+\sqrt{2(\lambda-a)T}\right)\right].$$

In particular,

$$\beta_0 = \lambda = r+\frac{1}{2}\sigma^2\xi^2,$$

and

$$\beta_r = \lambda - r = \frac{1}{2}\sigma^2\xi^2.$$

1.4 Closed-Form Bond Price

Combining the terms, the Black-Cox bond price is

$$B(V_0;V_B,T) = B_{\rm rf}(T) - q^{\xi}\left[\frac{\rho}{r}G_0(T)+\left(1-\frac{\rho}{r}-c\right)G_r(T)\right],$$

where

$$B_{\rm rf}(T) = \frac{\rho}{r}+\frac{r-\rho}{r}e^{-rT},$$
$$q = \frac{V_B}{V_0}, \qquad \xi = \frac{r-\delta}{\sigma^2}-\frac{1}{2},$$
$$L = \log\left(\frac{V_0}{V_B}\right), \qquad \alpha = \frac{2L^2}{\sigma^2},$$

and

$$\lambda = r+\frac{1}{2}\sigma^2\xi^2.$$

The function $G_a(T)$ is

$$G_a(T) = e^{-aT}\left[e^{\sqrt{\alpha(\lambda-a)}}\Phi\left(-\sqrt{\frac{\alpha}{2T}}-\sqrt{2(\lambda-a)T}\right)+e^{-\sqrt{\alpha(\lambda-a)}}\Phi\left(-\sqrt{\frac{\alpha}{2T}}+\sqrt{2(\lambda-a)T}\right)\right].$$

1.5 Credit Spread

Given the risky bond price $B(V_0; V_B, T)$, the risky yield y is defined implicitly by

$$B(V_0; V_B, T) = e^{-yT} + \int_0^T \rho e^{-yt}\, dt.$$

Equivalently,

$$B(V_0; V_B, T) = e^{-yT} + \frac{\rho}{y}(1 - e^{-yT}).$$

The credit spread is then

$$\mathrm{CS}(T) = y - r.$$

For a zero-coupon bond, if the risky price is $B_Z(T)$, then

$$B_Z(T) = e^{-(r+\mathrm{CS}(T))T},$$

so

$$\mathrm{CS}(T) = -\frac{1}{T}\log B_Z(T) - r.$$

1.6 Interpretation

The Black-Cox bond price has the form

$$B_{\text{risky}} = B_{\text{riskfree}} - \text{default adjustment}.$$

The default-free component is

$$B_{\text{rf}}(T) = \frac{\rho}{r} + \frac{r-\rho}{r}e^{-rT}.$$

The default adjustment depends on the distance to default,

$$q = \frac{V_B}{V_0},$$

the asset volatility σ, the payout rate δ, the recovery rate c, and the coupon rate ρ. A larger distance between V_0 and V_B reduces default risk, while a larger asset volatility σ increases the probability of hitting the default barrier. A higher recovery rate c increases the risky bond price.

In practice, one must calibrate or estimate

$$V_0, \quad V_B, \quad \sigma, \quad r, \quad \delta, \quad c, \quad \rho, \quad T.$$

A typical implementation uses equity market data to infer V_0 and σ, debt structure to estimate the default barrier V_B, and observed bond prices or CDS spreads to calibrate recovery and barrier parameters.